IF 最大平均值 IS 反向饱和电流

VR 最大瞬时值 VT 热电压（常温为mV）

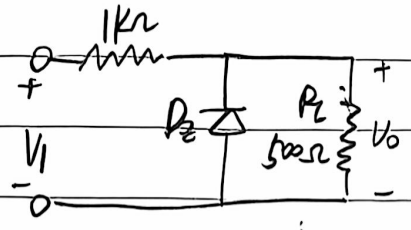
No.

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第二次作业

知识点2（除T4）

T1



$V_Z=6V$, $I_{Zmin}=5mA$, $I_{Zmax}=25mA$

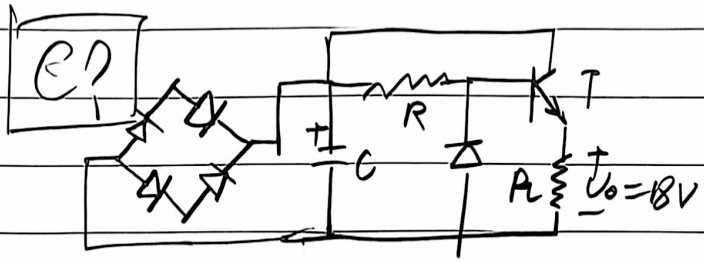
(1) $V_E=3V_Z=18V$

$V_I=10V$时, $V_O=\frac{10}{3}V$, $V_I=15V$时, $V_O=5V$, $V_I=35V$时, $V_O=6V$

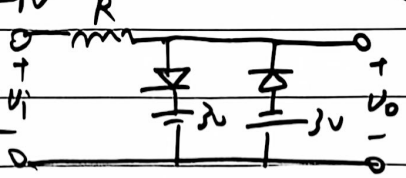
(2) $I=\frac{V_I-V_Z}{R}=24mA>I_{Zmax}$, 稳压管会被烧毁

T2 半波 全波 桥式

T3

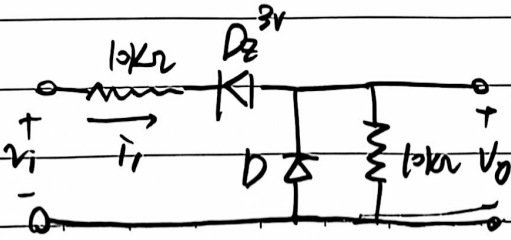


T5 1V, −1V



$V_i=5mtV$, $V_O=0.7V$

T6



$-10V\le V_i\le 0V$, $I_1=V_i\times10^{-4}A$, $V_o=0V$

$0\le V_i\le 3V$, $I_1=0A$, $V_O=0V$

$3V\le V_i\le 10V$, $I_1=\frac{V_i-3}{20}\times10^{-4}A$, $V_O=\frac{V_i-3}{2}V$

$-10V\le V_i\le 10V$

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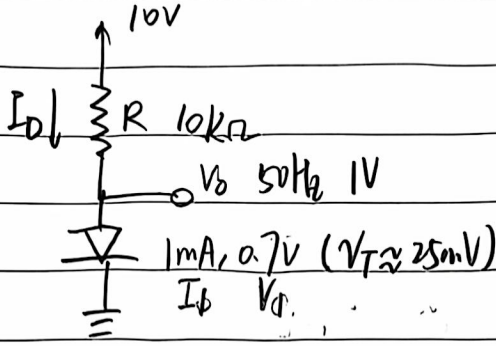
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$i = I_S e^{\frac{v}{V_T}}$

T7 D

T8



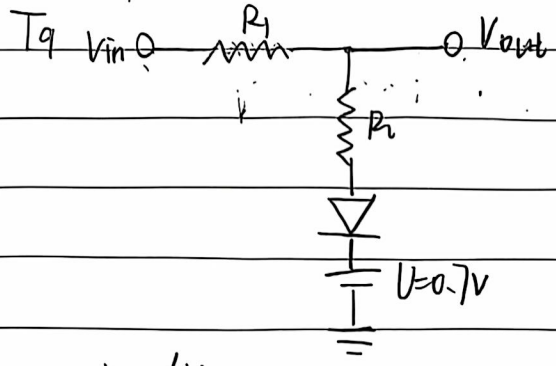
$$\begin{cases} \frac{10-V_2}{I_2} = 10k\Omega \\ \frac{I_2}{I_1} = e^{\frac{V_2-V_1}{V_T}} \end{cases} \Rightarrow \begin{cases} I_2 = 9.302\times10^{-3}A \\ V_2 = 0.698V \end{cases}$$

$\frac{V_1}{r}$

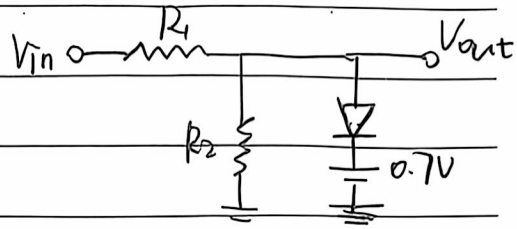
$$r_d = \frac{V_T}{i_d} = 26.88\Omega$$

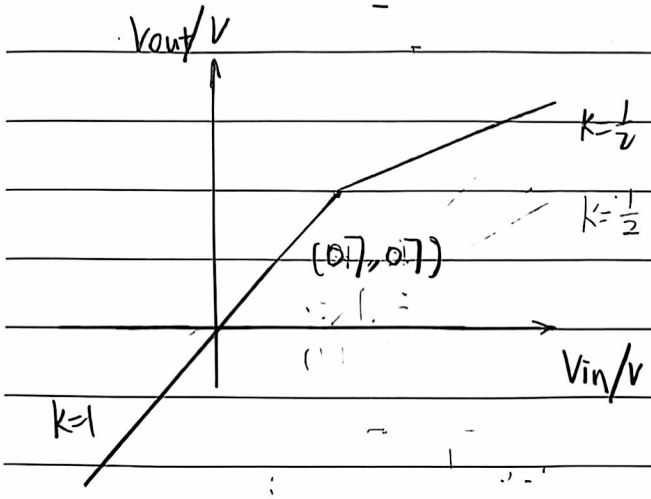
$$V = V_0 \frac{r_d}{R+r_d} = 2.68\times10^{-3}V$$

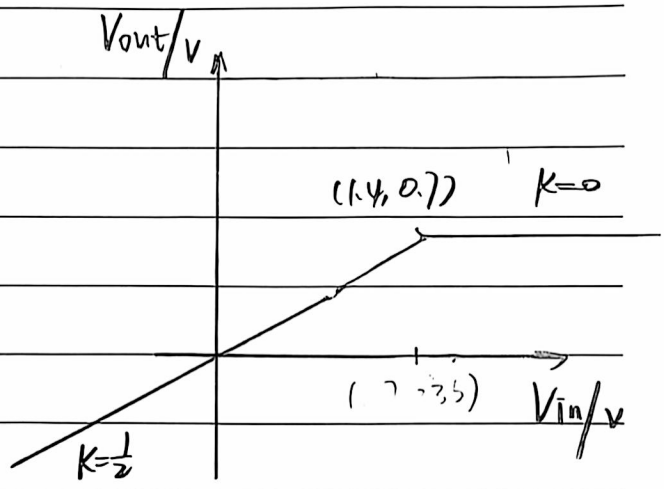
T9



T10





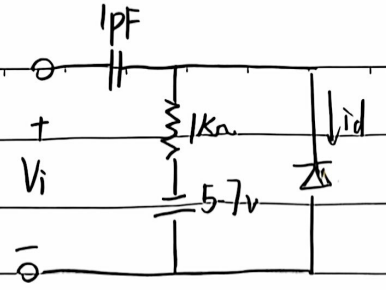


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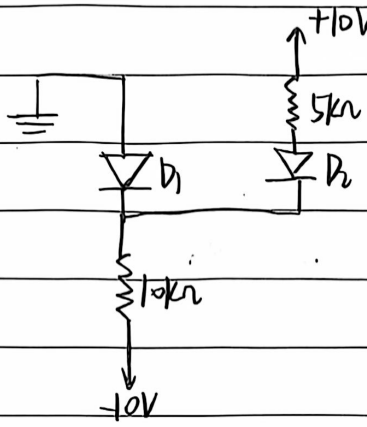
Date.

T11

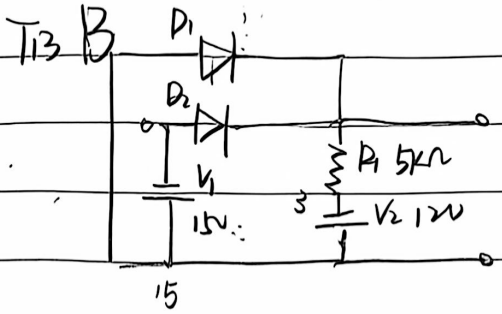


$i_d = 5.7mA \quad r = \frac{V_T}{I_d} \quad I = \frac{V_i}{r} = 2.28mA$

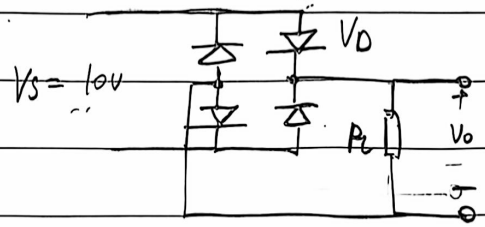
T12 C



T13 B

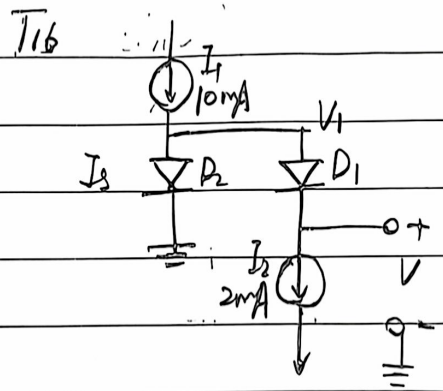


T14



$V = 12V_S - V_D = 13.44V$

T16



$$\begin{cases} I_S e^{\frac{V_1}{V_T}} = 8 \\ 10 I_S e^{\frac{V_1 - V}{V_T}} = 2 \end{cases} \Rightarrow V = V_T \ln 40 = 9.22mV$$

T15 整流器. 滤波器

$$\begin{cases} I_S e^{\frac{V_2}{V_T}} = 10 - I_2 \\ 10 I_S e^{\frac{V_2 - V_2}{V_T}} = I_2 \end{cases} \Rightarrow I_2 = \frac{10}{1 + \frac{1}{10} e^{\frac{V_2}{V_T}}}$$

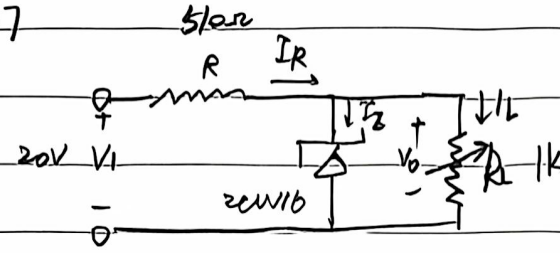
$I_2 = 5.75mA$

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T17



W=9V

P=250mW

$I_{min}=1mA$ $r_{max}=20\Omega$

(1) $I_R=\frac{20-9}{510}=21.6mA$

$I_L=\frac{9}{1000}=9mA$

$I_Z=I_R-I_L=12.6mA$

(2) $\Delta V_o=20\times0.2\times\frac{20\|1000}{20\|1000+510}=0.14V$

(3) $V_I=24V$

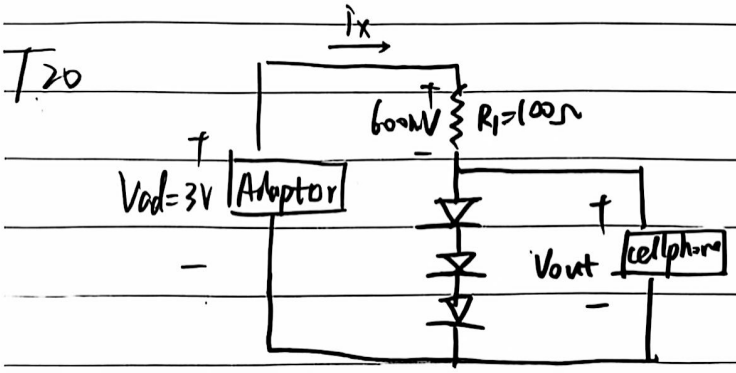
$I_Z=\frac{24-9}{510}-\frac{9}{RL}\leq 29.4mA$

$P=9\times29.4=264.6mW>250mW$ 超过允许值

T18 D

T19 B $i=C\frac{dv}{dt}$

T20



$V_D=800mV$

$V_T=25mV$

$i_x=\frac{3-3\times0.8}{100}=6mA$ $r=\frac{V_T}{i_x}=4.17\Omega$

$\Delta V_{out}=\Delta V_{ad}\cdot\frac{3r}{R+3r}=11.12mV$

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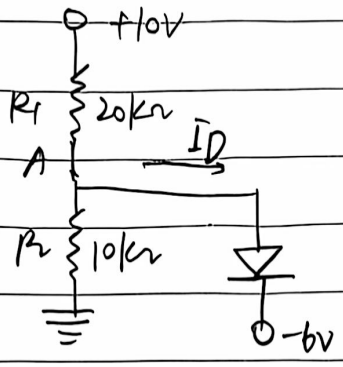
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硅 0.7 锗 0.3

T21 A

T22



$V_{on}=0.7V$

$V_A=-6+0.7=-5.3V$

$I_D=\frac{10+5.3}{20K}+\frac{5.3}{10K}=1.295mA$

三极管知识点

T1 正时电子和空穴同时导电

T2 少数

T3 [illegible]

T4 集电极 发射极 基极 PNP

T5 A

T6 A

T7 B

T8 D

